Thermodynamics Formulary

Author: Rosh Guadiana
Course: Thermodynamics
TA: Dr. Felipe Córdova Lozano
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First Law

$$w = -P_{\text{ext}}\Delta V$$

Second Law

Entropy originates here; it allows us to determine the direction of a reaction.

$$\Delta S = \frac{q_{\text{rev}}}{T}$$

$\Delta S_{\text{universe}} > 0$ then the process is non reversible, i.e. irreversible, spontaneous
$\Delta S_{\text{universe}} = 0$ the process is reversible where $\Delta S_{\text{universe}} = \Delta S_{\text{syst.}} + \Delta S_{\text{surr.}}$
Irreversible isothermal: $\Delta S_{\text{surr.}} = -\frac{q_{\text{sys}}}{T}$ $\Delta u = 0$ $-w = q$

General Entropy Changes

For any reversible process of an ideal gas, derived from $dU = TdS - PdV$:

Simultaneous change of Volume and Temperature:

$$\Delta S = nC_{v,m}\ln\left(\frac{T_f}{T_i}\right) + nR\ln\left(\frac{V_f}{V_i}\right)$$

Simultaneous change of Pressure and Temperature:

$$\Delta S = nC_{p,m}\ln\left(\frac{T_f}{T_i}\right) - nR\ln\left(\frac{P_f}{P_i}\right)$$

A gas expands isothermally

The reversible work in an **isothermal expansion** is $w_{\text{rev}} = -nRT\ln\left(\frac{V_f}{V_i}\right)$, hence because for this isothermal process of a perfect gas $q = -w$, we have:

$$\Delta S = nR\ln\left(\frac{V_f}{V_i}\right)$$

Here the change is isothermal so we could use $P_iV_i = P_fV_f$.
We could also use Hess's law to calculate the entropy ΔS.
Enthalpy in physical process: $\Delta H = nC_{p,m}$

If expansion is irreversible (spontaneous) then $\Delta S_{\text{universe}} > 0$ where $\Delta S_{\text{universe}} = \Delta S_{\text{system}} + \Delta S_{\text{surroundings}}$.

Change in entropy by heating

$$\Delta S = nC_{p,m} \ln\left(\frac{T_2}{T_1}\right)$$

where $T_2 > T_1$.
Also

$$\Delta S = m\mathbb{S} \ln\left(\frac{T_2}{T_1}\right)$$

where $T_2 > T_1$

Adiabatic process $q = 0$

Work of reversible adiabatic expansion:

$$w_{\mathrm{ad}} = nC_{v,m}\Delta T = \Delta U$$

In a monoatomic gas $C_v = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$. In a diatomic gas $C_v = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$.

Temperature change, reversible adiabatic expansion:

$$T_f = T_i \left(\frac{V_i}{V_f}\right)^{\frac{1}{c}}$$

where $c = \frac{C_{v,m}}{R}$

Pressure change, reversible adiabatic expansion:

$$P_f V_f^{\gamma} = P_i V_i^{\gamma}$$

where $\gamma = \frac{C_{p,m}}{C_{v,m}}$

$$T_i V_i^{\gamma-1} = T_f V_f^{\gamma-1}$$

Avogadro's Principle / Entropy of Mixing

When two ideal gases mix isothermally and isobarically, the process is spontaneous, driven entirely by an increase in entropy. Using mole fractions for a mixture of gas A and B: $X_A = \frac{n_A}{n_A+n_B}$ and $X_B = \frac{n_B}{n_A+n_B}$.

$$\Delta S_{\mathrm{mix}} = -R(n_A \ln(X_A) + n_B \ln(X_B))$$

Because a mole fraction (X) is always between 0 and 1, its natural logarithm is negative. Multiplying by $-R$ guarantees that $\Delta S_{\mathrm{mix}} > 0$.

Carnot Cycle

We have 4 processes, four reversible stages in which a gas (the working substance) is either expanded or compressed in a particular sequence of ways.

$$\Delta U_{\text{cycle}} = \Delta S_{\text{cycle}} = \Delta H_{\text{cycle}} = 0$$

Efficiency e

The efficiency is always positive:

$$e = 1 + \frac{q_3}{q_1}$$

$$e = 1 - \frac{T_C}{T_H}$$

$$e = -\frac{w_{\text{total}}}{q_1}$$

Solving for Carnot (Explicit Steps)

- **Stage 1: Isothermal Reversible Expansion** (at T_H) Heat is transferred from a hot source to the gas.

$$\Delta U_1 = 0$$

$$q_1 = -w_1 = nRT_H \ln\left(\frac{V_2}{V_1}\right)$$

$$\Delta S_1 = \frac{q_1}{T_H} = nR\ln\left(\frac{V_2}{V_1}\right)$$

- **Stage 2: Adiabatic Reversible Expansion** (T_H to T_C) The temperature falls from T_H to T_C.

$$q_2 = 0$$

$$\Delta U_2 = w_2 = nC_{v,m}(T_C - T_H)$$

$$\Delta S_2 = 0$$

- **Stage 3: Isothermal Reversible Compression** (at T_C) Energy as heat is transferred from the gas to a cold sink.

$$\Delta U_3 = 0$$

$$q_3 = -w_3 = nRT_C \ln\left(\frac{V_4}{V_3}\right)$$

$$\Delta S_3 = \frac{q_3}{T_C} = nR\ln\left(\frac{V_4}{V_3}\right)$$

- **Stage 4: Adiabatic Reversible Compression** (T_C to T_H) Restores the system to its initial state.

$$q_4 = 0$$

$$\Delta U_4 = w_4 = nC_{v,m}(T_H - T_C)$$

$$\Delta S_4 = 0$$

Total Cycle Outcomes: Because the cycle returns to its initial state, all state functions have a net change of zero:

$$\Delta U_{\text{total}} = \Delta U_1 + \Delta U_2 + \Delta U_3 + \Delta U_4 = 0$$

$$\Delta S_{\text{total}} = \Delta S_1 + \Delta S_2 + \Delta S_3 + \Delta S_4 = 0$$

$$w_{\text{total}} = w_1 + w_2 + w_3 + w_4$$

Stages of a reaction

i.e. argon changes temp and volume, we could calculate the entropy of each step separately. When volume changes:

$$\Delta S = nR \ln\left(\frac{V_f}{V_i}\right)$$

When temp changes:

$$\Delta S = nC_{v,m} \ln\left(\frac{T_f}{T_i}\right)$$

Temperature exchange of two substances: We calculate $T_{\text{equilibrium}}$ and then we know $\Delta S = \Delta S_1 + \Delta S_2$, we sum both changes.

Phase changes in a reaction

For example, vaporization of water. This is a reversible process. Assuming **constant pressure**, we know $q = \Delta H_{\text{vap}}$. Thus:

$$\Delta S = \frac{q}{T} = \frac{\Delta H_{\text{vap}}}{T}$$

This assumption is useful in reactions, when pressure is constant

$$\Delta S = \frac{\Delta H}{T}$$

Unit Conversions

Pressure

$$1 \text{ atm} = 101325 \text{ Pa} = 101.325 \text{ kPa} = 1.01325 \text{ bar}$$

$$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mmHg}$$

$$1 \text{ bar} = 10^5 \text{ Pa} = 0.986923 \text{ atm}$$

Energy & Work

$$1 \text{ J} = 1 \text{ kg} \cdot \frac{\text{m}^2}{\text{s}^2} = 1 \text{ V} \cdot \text{C}$$

$$1 \text{ cal} = 4.184 \text{ J (Thermochemical calorie)}$$

$$1 \text{ L} \cdot \text{atm} = 101.325 \text{ J}$$

$$1 \text{ eV} \approx 1.602176634 \times 10^{-19} \text{ J}$$

Temperature

$$T(K) = T^\circ C + 273.15$$

$$T^\circ C = (T^\circ F - 32) \times \frac{5}{9}$$

$$T^\circ R = T^\circ F + 459.67 (\text{Rankine, absolute scale})$$

$$T^\circ R = \frac{9}{5} T(K)$$

Volume

$$1\text{m}^3 = 1000 \text{ L} = 10^6 \text{cm}^3$$

$$1000\text{cm}^3 = 1L$$

$$1 \text{ L} = 1\text{dm}^3 = 1000 \text{ mL} = 1000\text{cm}^3$$

Fundamental Constants

Constant	**Symbol**	**Value**
Universal Gas Constant	R	$8.314462618 \frac{\text{J}}{\text{mol} \cdot \text{K}}$ $0.082057366 \text{ L} \cdot \frac{\text{atm}}{\text{mol} \cdot \text{K}}$ $1.987204258 \frac{\text{cal}}{\text{mol} \cdot \text{K}}$
Avogadro's Number	N_A	$6.02214076 \times 10^{23} \text{mol}^{-1}$
Standard Atm Pressure	P_{atm}	101325 Pa